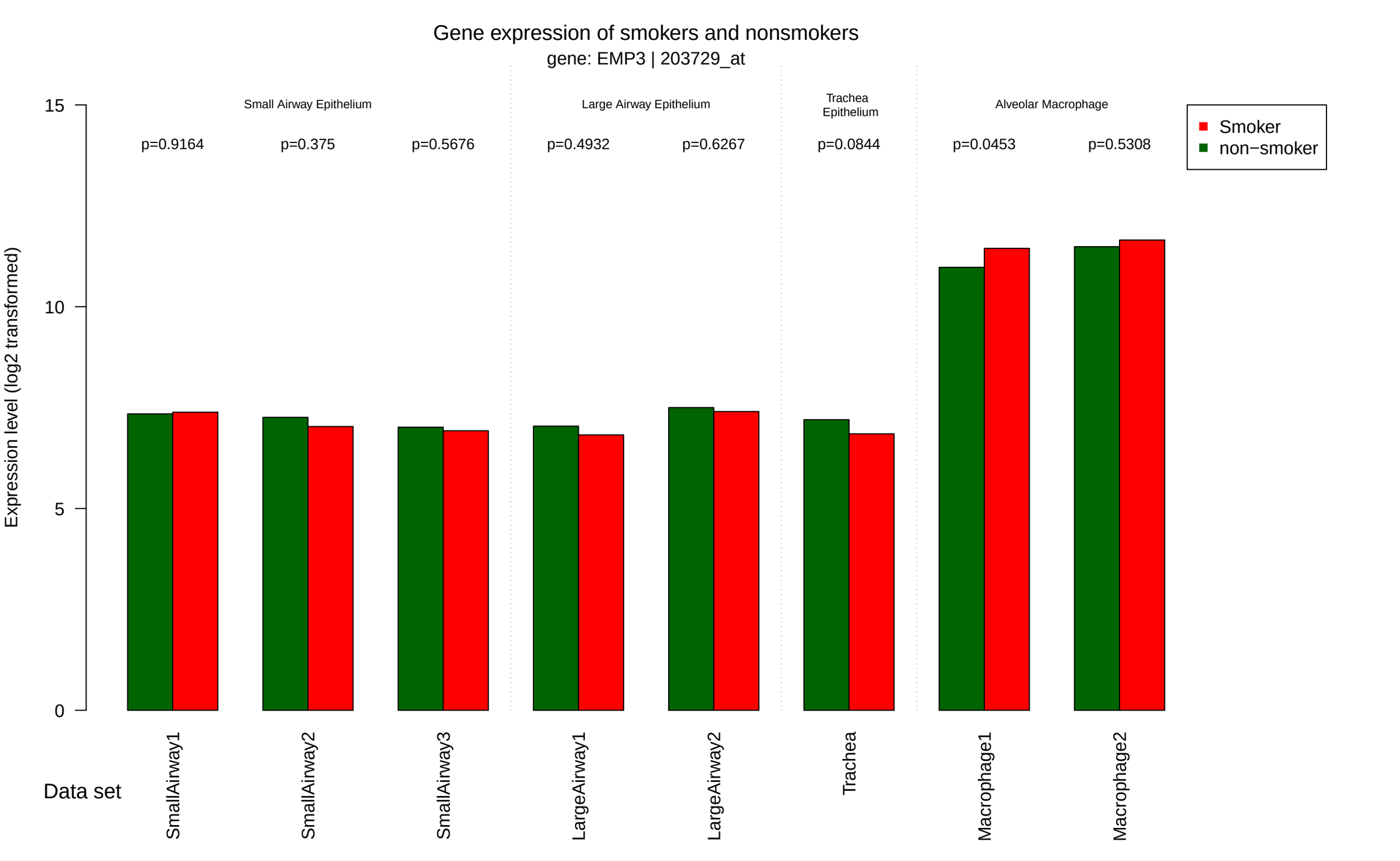
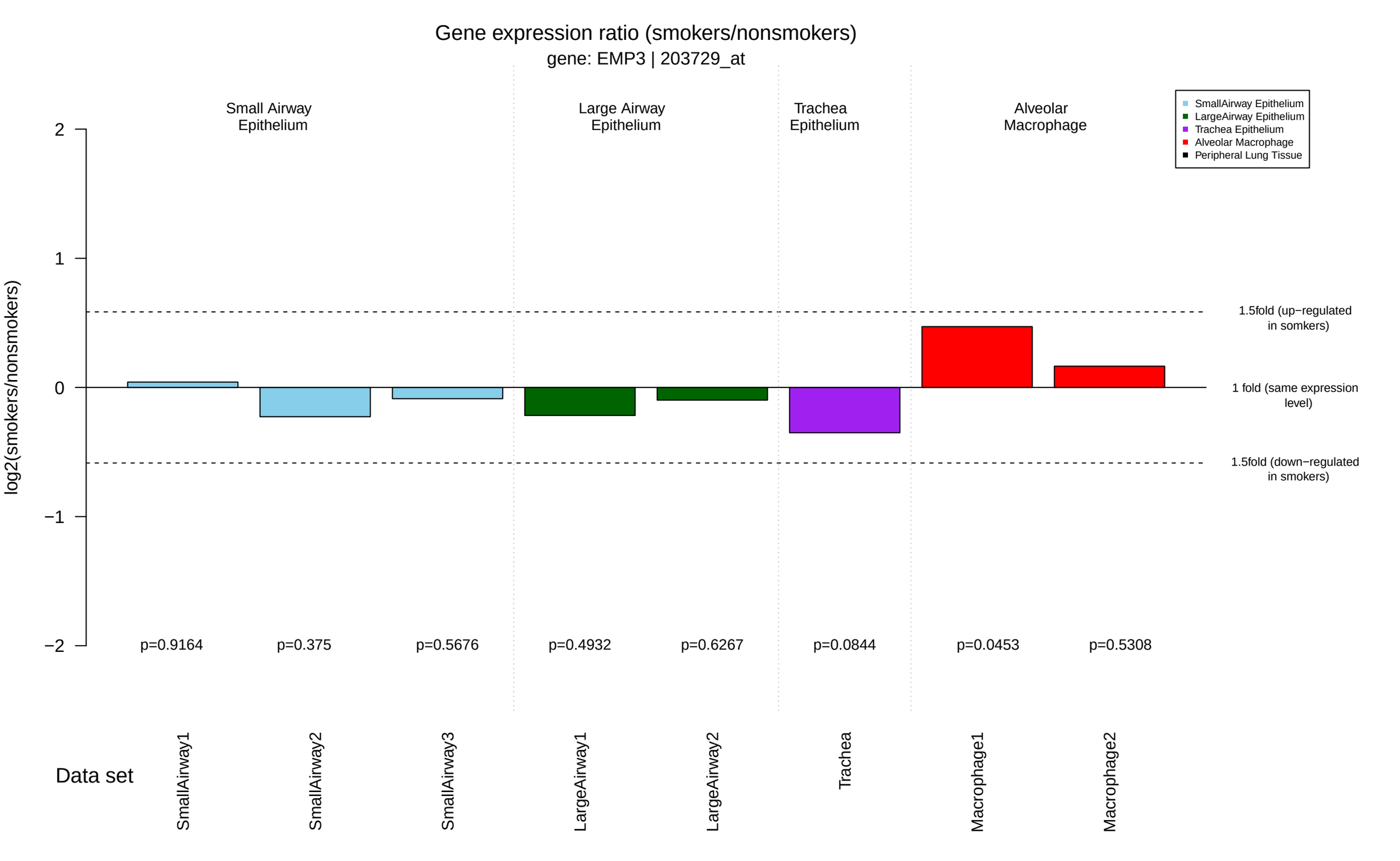
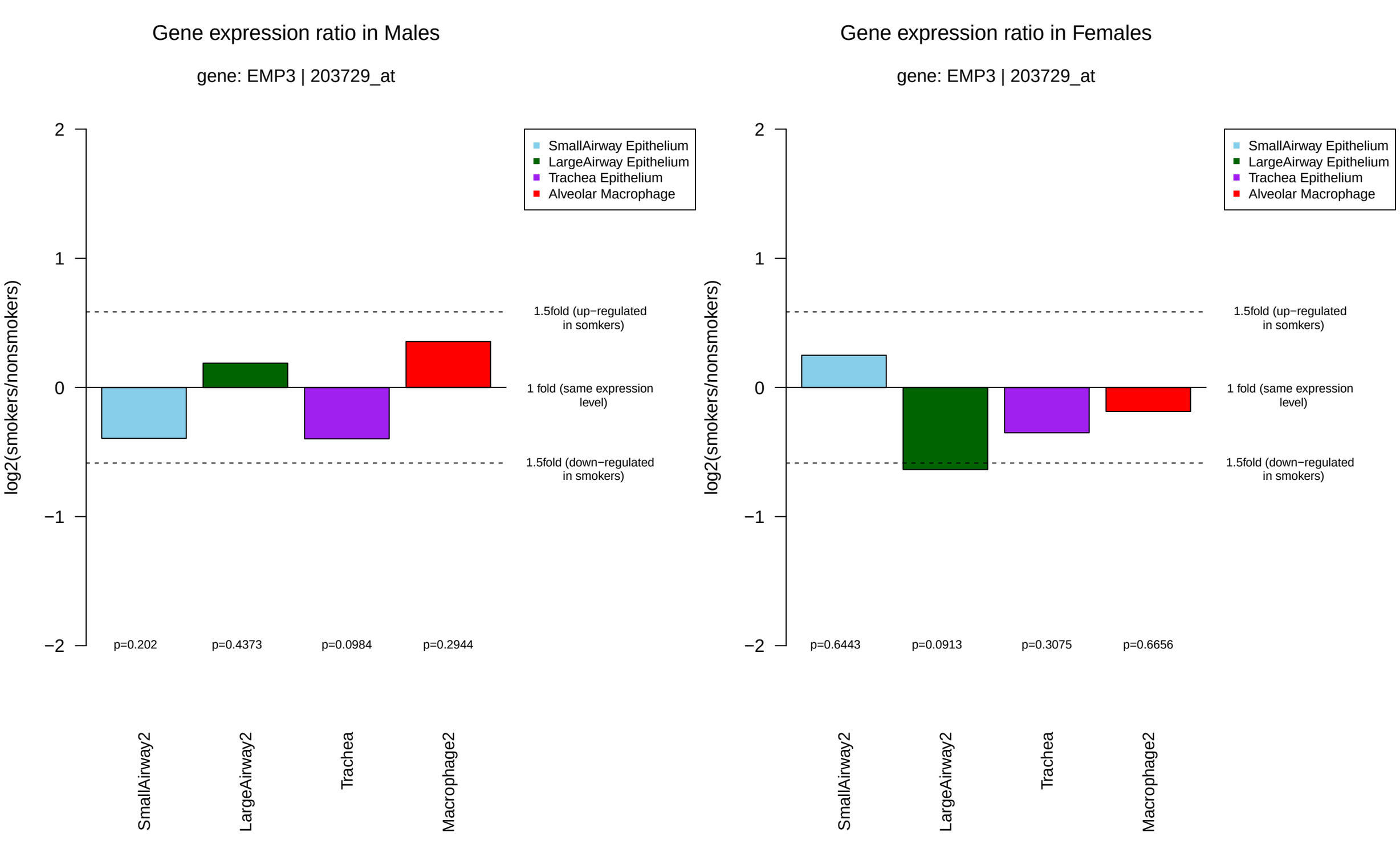
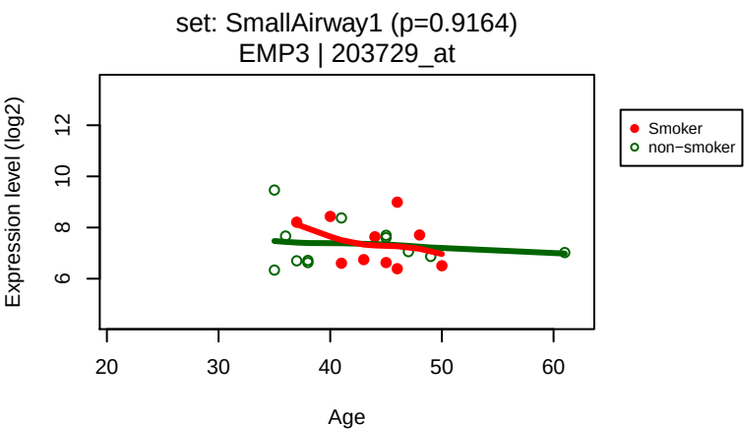
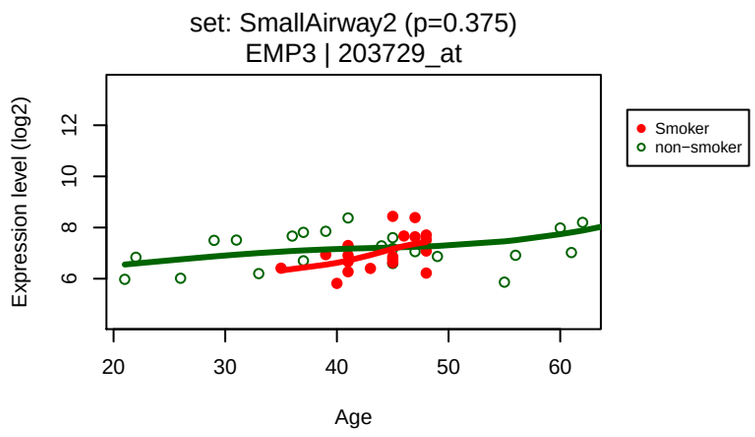
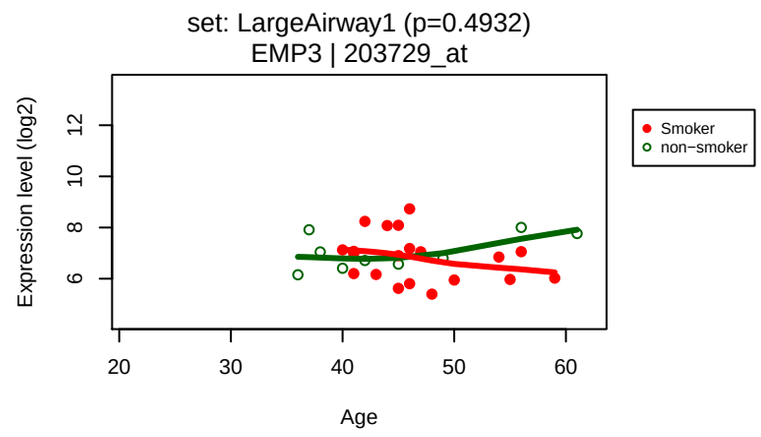

Gene expression of smokers and nonsmokers
gene: EMP3 | 203729_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9164
p=0.375
p=0.5676
p=0.4932
p=0.6267
p=0.0844
p=0.0453
p=0.5308
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

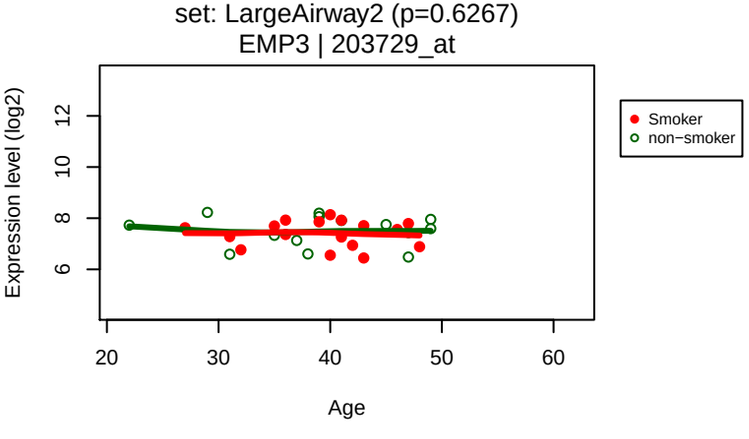

Gene expression ratio (smokers/nonsmokers)
gene: EMP3 | 203729_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9164
p=0.375
p=0.5676
p=0.4932
p=0.6267
p=0.0844
p=0.0453
p=0.5308
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

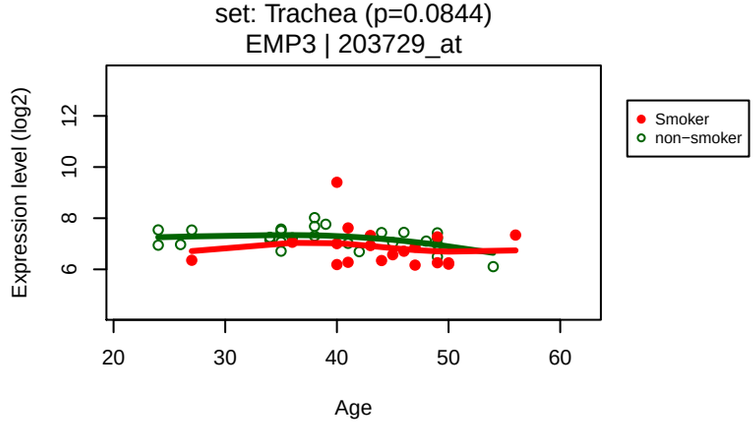

Gene expression ratio in Males
gene: EMP3 | 203729_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.202
p=0.4373
p=0.0984
p=0.2944
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EMP3 | 203729_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6443
p=0.0913
p=0.3075
p=0.6656
SmallAirway2
LargeAirway2
Trachea
Macrophage2

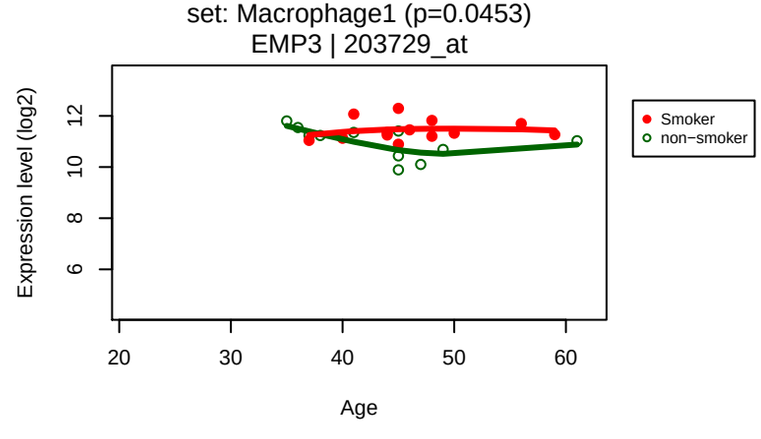

set: SmallAirway1 (p=0.9164)
EMP3 | 203729_at
Expression level (log2)
Age
Smoker
non−smoker

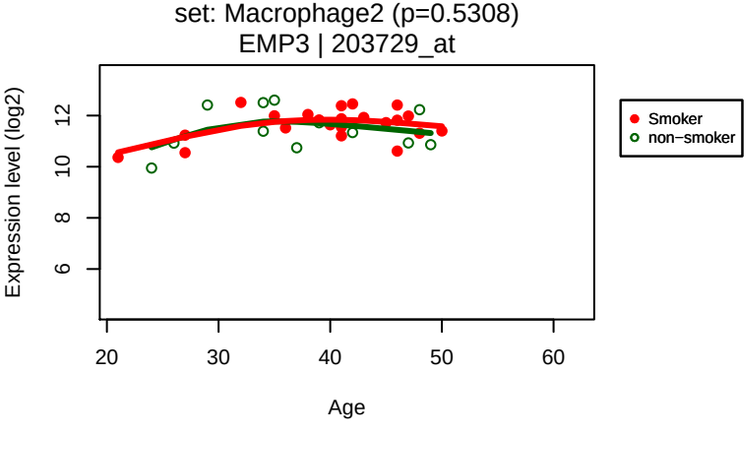

set: SmallAirway2 (p=0.375)
EMP3 | 203729_at
Expression level (log2)
Age
Smoker
non−smoker

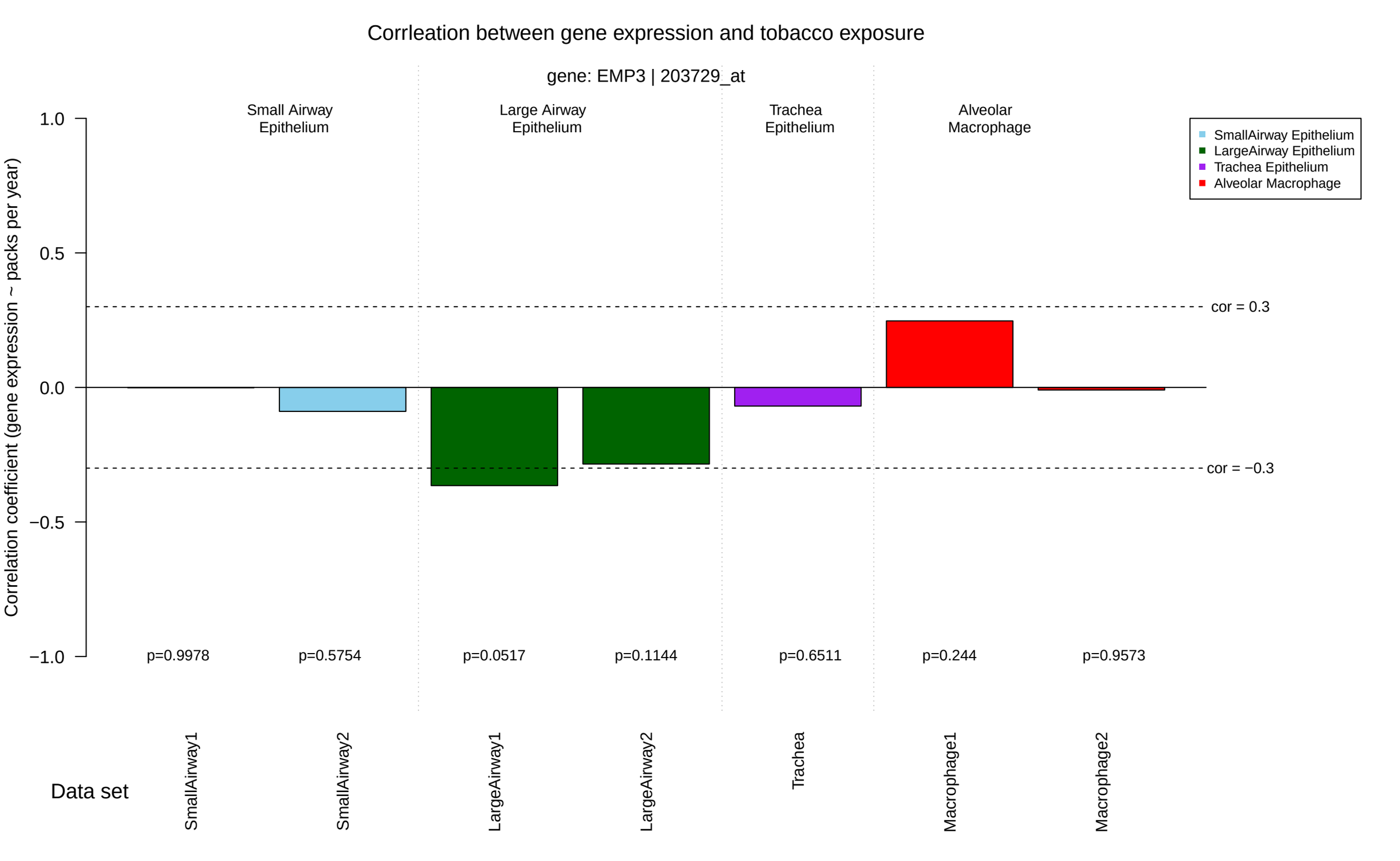

set: LargeAirway1 (p=0.4932)
EMP3 | 203729_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6267)
EMP3 | 203729_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0844)
EMP3 | 203729_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0453)
EMP3 | 203729_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5308)
EMP3 | 203729_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EMP3 | 203729_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9978
p=0.5754
p=0.0517
p=0.1144
p=0.6511
p=0.244
p=0.9573
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2